Part Three - Make it yours

Last meeting you got one full loop playing. Today it becomes yours.

Start with five minutes of play-testing. Let a neighbor play your game, and keep your hands off the keyboard. Watching somebody else play is the fastest way to find out what's broken.

Step 4 - Replace the placeholders

Your turn. Your character, your collectibles, your hazards, your world. Every color, every name, every shape the starter marks `CHANGE`. Space, ocean, forest, city, a dream. Whatever your sentence from 7.1 said.

Step 5 - Your screens, in your words

Your turn. Rewrite the win and lose screens. *"YOU WIN!"* is the starter talking. Make it sound like you, and like your game.

Keep the `textAlign(LEFT, BASELINE);` at the end of each one, as in Chapter 6.

Step 6 - Tune it

Your turn. Play it. Too easy? Faster hazards, or a higher `goal`. Too hard? Slower hazards, or a bigger hitbox in your `super(...)` line.

You built your game, so you're the best player it'll ever have. That means it's probably too hard for everybody else. Your neighbor is the one to tune it for.

7.5 Three coins, three `if`s

The starter has three coins and two hazards. Each one has its own variable and its own `if`. Yes, that's repetitive. It's also code you can read from top to bottom and change one piece at a time. That's a trade-off, and for a first game it's the right one.

Want a fourth coin? It goes in four places: declare `coin4`, create it in `restartGame()`, move and draw it in `playGame()`, and give it an `if`.

> **Watch out.** Copy the `coin3` block to make `coin4`, and every `coin3` in the copy has to become `coin4`. Miss the one in `moveTo()`, and touching `coin4` moves `coin3` away. `coin4` sits still under your player, scoring a point every frame, and you win the whole game in under a second. Read the copied lines out loud, name by name.

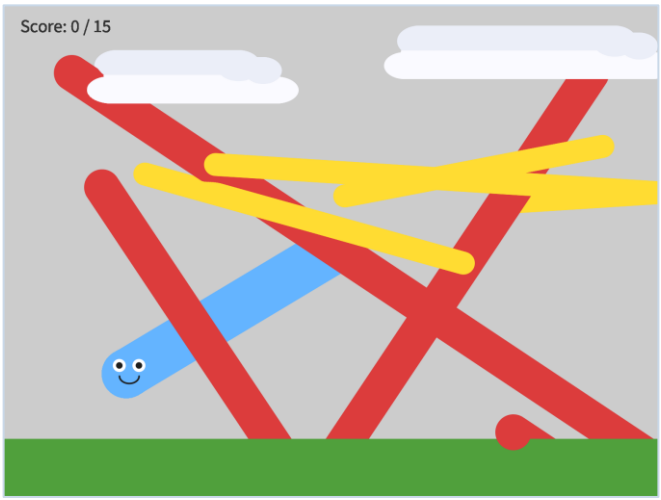
Want twenty coins? That's what Bonus D is for.

Step 7 - One thing that's your idea

Your turn. Add one thing that's your idea. Some places to start:

- **Lives**, from Chapter 6, in place of instant game over.
- **Hazards that speed up** as the score climbs.
- **A special ability** on a key: a speed burst, a shield, a teleport. It goes in `keyPressed()`.
- **A custom hazard**, in its own `extends Blob` tab: a shark, a storm cloud, a meteor.
- **A title screen** before the game starts. That's a fourth mode.
- **Bouncing.** One line in `setup()`, right before `restartGame()`:

```
  Sprite.screenBehavior = ScreenBehavior.BouncyWalls;   // everything bounces
off the edges
```



`BouncyWalls`, with `background()` switched off to show the paths. The hazards bounce off the bottom edge, behind the ground, and the coins bounce off the right.

It changes every sprite in your game at once. That's why the clouds bounce too.

> **Argue it.** Play your game with bouncing on, then with it off. It's the same code, and a different game. Which one is better? Write down which one you kept, and why.

- **A bonus chapter.** Pictures for your character (Bonus A), sounds (Bonus B), an animated character (Bonus C), or lots of coins (Bonus D). They're at the back of the book.

Your turn. Optional: a second level. When the player wins, rebuild the world harder, with faster hazards, a different background, and a higher goal, and keep the score. That's `restartGame()` with a level number, and it turns a demo into a game people want to keep playing.